

## DBMS -- Functional Dependency

Target: Explain trivial vs non-trivial FD in 20s. State all 3 Armstrong's rules by name.

What Is Functional Dependency?

A Functional Dependency (FD) is a constraint between two sets of attributes in a relation.

If two tuples (rows) have the same values for attributes A1, A2, ... then they must also have the same values for attributes B1, B2, ...

Notation:  $X \twoheadrightarrow Y$  (read as: "X functionally determines Y")

Meaning: Knowing the value of X is enough to determine the value of Y.

Example:

StudentID  $\twoheadrightarrow$  StudentName

If two rows have the same StudentID, they MUST have the same StudentName.

StudentID  $\twoheadrightarrow$  {Name, Age, Course}

Knowing StudentID alone tells you Name, Age, and Course.

### Types of Functional Dependency

mermaid

graph TD

FD[Functional Dependency  $X \twoheadrightarrow Y$ ]

FD --> T[Trivial FD]

FD --> NT[Non-Trivial FD]

NT --> CNT[Completely Non-Trivial FD]

#### 1. Trivial FD

Rule:  $X \twoheadrightarrow Y$  is trivial if Y is a subset of X.

Y is already "inside" X -- knowing X obviously tells you Y. Always holds. Not useful for normalization.

Examples:

{StudentID, Name}  $\twoheadrightarrow$  {StudentID} trivial (StudentID is already in the left side)

{StudentID, Name}  $\twoheadrightarrow$  {Name} trivial

{A, B}  $\twoheadrightarrow$  {A} trivial

{A}  $\twoheadrightarrow$  {A} trivial

Trivial FDs always hold in every relation -- they don't tell you anything new.

#### 2. Non-Trivial FD

Rule:  $X \twoheadrightarrow Y$  is non-trivial if Y is NOT a subset of X.

Y contains something that X doesn't. This is the useful kind for normalization.

Examples:

StudentID  $\twoheadrightarrow$  StudentName non-trivial (Name is not part of StudentID)

StudentID  $\twoheadrightarrow$  {Name, Age} non-trivial

{EmpID}  $\twoheadrightarrow$  {DeptName} non-trivial

#### 3. Completely Non-Trivial FD

Rule:  $X \twoheadrightarrow Y$  is completely non-trivial if  $X \cap Y = \emptyset$  (X and Y share no attributes at all).

The most "pure" kind of FD -- the left and right sides have nothing in common.

Examples:

StudentID  $\twoheadrightarrow$  StudentName completely non-trivial (no overlap)

{EmpID}  $\twoheadrightarrow$  {DeptID, Salary} completely non-trivial

Contrast:

{StudentID, Name}  $\twoheadrightarrow$  {Name, Age}

$X \twoheadrightarrow Y = \{Name\}$  NOT completely non-trivial (Name appears on both sides)

But IS non-trivial (Age is not in X)

Summary Table

Type

Condition

Always holds?

Useful?

Trivial  
 $Y \rightarrow X$   
[OK] Always  
No new info

Non-Trivial  
 $Y \rightarrow X$   
Depends on data  
[OK] Yes

Completely Non-Trivial  
 $X \rightarrow Y =$   
Depends  
[OK] Most useful

### Armstrong's Axioms (Inference Rules)

Armstrong's Axioms are a complete and sound set of rules for deriving all functional dependencies from a given set.

"If F is a set of functional dependencies, then the closure of F (written  $F^+$ ) is the set of all FDs logically implied by F."

There are 3 rules (axioms):

Rule 1 -- Reflexivity Rule  
If  $Y \rightarrow X$ , then  $X \rightarrow Y$   
If Y is a subset of X, then X functionally determines Y.  
(This generates all trivial FDs.)  
Examples:  
 $\{StudentID, Name\} \rightarrow \{StudentID\}$  reflexivity  
 $\{A, B, C\} \rightarrow \{B, C\}$  reflexivity  
 $\{A\} \rightarrow \{A\}$  reflexivity

Rule 2 -- Augmentation Rule  
If  $X \rightarrow Y$ , then  $XZ \rightarrow YZ$  (for any set of attributes Z)  
If X determines Y, then adding the same attribute Z to both sides still holds.  
Examples:

$StudentID \rightarrow Name$   
 $\{StudentID, Age\} \rightarrow \{Name, Age\}$  augmented with Age  
 $EmpID \rightarrow Salary$   
 $\{EmpID, DeptID\} \rightarrow \{Salary, DeptID\}$  augmented with DeptID

Think of it as: "Adding the same thing to both sides doesn't break the dependency."

Rule 3 -- Transitivity Rule  
If  $X \rightarrow Y$  and  $Y \rightarrow Z$ , then  $X \rightarrow Z$   
If X determines Y, and Y determines Z, then X determines Z.  
(Just like transitivity in math: if  $a=b$  and  $b=c$  then  $a=c$ .)  
Examples:

$StudentID \rightarrow DeptID$   
 $DeptID \rightarrow DeptName$   
 $StudentID \rightarrow DeptName$  transitivity  
 $EmpID \rightarrow ManagerID$   
 $ManagerID \rightarrow ManagerName$   
 $EmpID \rightarrow ManagerName$  transitivity

This is the basis of transitive dependency -- which is exactly what 3NF aims to remove!

Armstrong's Rules -- Quick Reference

| Rule | Statement | Short form |
|------|-----------|------------|
| 1    | ...       | ...        |
| 2    | ...       | ...        |
| 3    | ...       | ...        |
| 4    | ...       | ...        |
| 5    | ...       | ...        |
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| 98   | ...       | ...        |
| 99   | ...       | ...        |
| 100  | ...       | ...        |

## Reflexivity

If  $Y \subseteq X$ , then  $X \subseteq Y$

## Subsets are always determined

## Augmentation

If  $X \leq Y$ , then  $XZ \leq YZ$

Adding same attrs to both sides is valid

## Transitivity

If  $X \sim Y$  and  $Y \sim Z$ , then  $X \sim Z$

## FDs chain like equations

These 3 rules are sound (only generate valid FDs) and complete (can generate ALL valid FDs).

## FD and Normalization -- The Connection

...

Trivial FD      always present, not a problem

Non-Trivial FD the ones we care about in normalization

|                    |                                                   |              |
|--------------------|---------------------------------------------------|--------------|
| Partial dependency | non-key attribute depends on PART of composite PK | violates 2NF |
|--------------------|---------------------------------------------------|--------------|

Transitive dependency non-key A non-key B violates 3NF

(discovered via Armstrong's Transitivity rule)

...

## Your 40-Second Script -- Functional Dependency

"A functional dependency  $X \twoheadrightarrow Y$  means knowing  $X$  uniquely determines  $Y$ . If  $Y$  is a subset of  $X$  it's trivial -- always holds, not useful. If  $Y$  is not a subset of  $X$  it's non-trivial -- these are the meaningful ones for normalization. Armstrong's three axioms let us derive all FDs: Reflexivity says subsets are always determined, Augmentation says you can add the same attributes to both sides, and Transitivity says if  $X \twoheadrightarrow Y$  and  $Y \twoheadrightarrow Z$  then  $X \twoheadrightarrow Z$  -- which is exactly what causes transitive dependencies in 3NF violations."

### Follow-Up Questions (Expect These)

Q: What is the closure of a set of attributes?

The closure of X (written  $X^+$ ) is the set of all attributes that X can functionally determine using the given FDs. Used to check if X is a candidate key (if  $X^+ = \text{all attributes}$ , X is a superkey).

Q: How does transitive dependency relate to Armstrong's Transitivity rule?

Transitive dependency IS Armstrong's Transitivity applied to non-key attributes. If PK  $A \rightarrow B$  and  $A \rightarrow C$  (where A is not a key), then PK  $B \rightarrow C$  via transitivity -- that B column should be moved out (3NF fix).

Q: What is the difference between trivial and non-trivial FD in one line?

Trivial: Y is already inside X (adds no info). Non-Trivial: Y contains something new that X doesn't have.